

Day 4 — Wazuh Indexing, Dashboards & File Integrity Monitoring

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Objective

To configure and validate File Integrity Monitoring (FIM) on a Linux agent, enable and assess Wazuh's Vulnerability Detector, and build a custom SOC dashboard with multiple visualizations exportable as a client-ready artefact.

System Information

Detail	Value
Wazuh Manager	Ubuntu Server 22.04 LTS — 192.168.56.104
FIM Target Agent	Kali Linux (agent-kali)
Dashboard Build Environment	OpenSearch Dashboards (via Wazuh Dashboard)

Part 1 — File Integrity Monitoring (FIM)

Configuration

FIM was verified as already active by default on the Kali agent, monitoring `/etc`, `/bin`, `/usr/bin`, `/sbin`, and `/boot` with full verification options (size, permissions, owner, group, mtime, inode, and cryptographic hashes MD5/SHA1/SHA256). The scan frequency was temporarily reduced from 43200 seconds (12 hours) to 300 seconds (5 minutes) to allow for faster testing and verification.,

- `<frequency>300</frequency>`

```
(fara@kali)-[~/Desktop]
$ sudo nano /var/ossec/etc/ossec.conf
```

```

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>

  <!-- Files/directories to ignore -->
  <ignore>/etc/mtab</ignore>
  <ignore>/etc/hosts.deny</ignore>
  <ignore>/etc/mail/statistics</ignore>
  <ignore>/etc/random-seed</ignore>
  <ignore>/etc/random.seed</ignore>
  <ignore>/etc/adjtime</ignore>
  <ignore>/etc/httpd/logs</ignore>
  <ignore>/etc/utmpx</ignore>
  <ignore>/etc/wtmpx</ignore>
  <ignore>/etc/cups/certs</ignore>
  <ignore>/etc/dumpdates</ignore>
  <ignore>/etc/svc/volatile</ignore>

  <!-- File types to ignore -->
  <ignore type="sregex">.log$|.swp$</ignore>

  <!-- Check the file, but never compute the diff -->
  <nodiff>/etc/ssl/private.key</nodiff>

  <skip_nfs>yes</skip_nfs>
  <skip_dev>yes</skip_dev>
  <skip_proc>yes</skip_proc>
  <skip_sys>yes</skip_sys>

  <!-- Nice value for Syscheck process -->
  <process_priority>10</process_priority>

  <!-- Maximum output throughput -->
  <max_eps>50</max_eps>

  <!-- Database synchronization settings -->
  <synchronization>
    <enabled>yes</enabled>
    <interval>5m</interval>
    <max_eps>10</max_eps>
  </synchronization>
</syscheck>
```

Test and Verification

A controlled modification was made to /etc/passwd by appending a test comment line and The Wazuh agent was restarted to force an immediate scan:

- `sudo bash -c 'echo "# FIM test entry - Day 4" >> /etc/passwd'`
- `sudo /var/ossec/bin/wazuh-control restart`

```
(fara@kali)-[~/Desktop]
$ sudo bash -c 'echo "# FIM test entry - Day 4" >> /etc/passwd'

(fara@kali)-[~/Desktop]
$ tail -3 /etc/passwd
stunnel:x:971:971:stunnel service system account:/var/run/stunnel:
/usr/sbin/nologin
wazuh:x:119:125::/var/ossec:/sbin/nologin
# FIM test entry - Day 4

(fara@kali)-[~/Desktop]
$ sudo /var/ossec/bin/wazuh-control restart
Killing wazuh-modulesd ...
Killing wazuh-logcollector ...
Killing wazuh-syscheckd ...
Killing wazuh-agentd ...
Killing wazuh-execd ...
Wazuh v4.9.0 Stopped
Starting Wazuh v4.9.0 ...
Started wazuh-execd ...
Started wazuh-agentd ...
Started wazuh-syscheckd ...
Started wazuh-logcollector ...
Started wazuh-modulesd ...
Completed.
```

The modification was successfully detected and an alert was generated and confirmed in the Wazuh Dashboard under Security Events, correctly identifying the change to /etc/passwd

Wazuh Dashboard interface showing the Events tab. The search filters are set to: manager.name: wazuh.manager, rule.groups: syscheck, agent.id: 001. The results show 1 hit for the event "Integrity check..." on Jul 2, 2026, at 17:17:43.361, related to agent-kali.

timestamp	agent.name	syscheck.p...	syscheck.e...	rule.descrip...	rule.level	rule.id
Jul 2, 2026 @ ...	agent-kali	/etc/passwd	modified	Integrity check...	7	550

The test change was subsequently reverted to restore the file to its original state:

- `sudo sed -i '/# FIM test entry - Day 4/d' /etc/passwd`

Outcome

FIM was confirmed to be actively monitoring critical system directories and correctly generating real-time alerts on unauthorized file modifications, a core detection capability for identifying attacker tampering with system files such as /etc/passwd.

Part 2 — Vulnerability Detector

Configuration Status

The Vulnerability Detector module was confirmed enabled by default on the Wazuh Manager:

```
Session Actions Edit View Help
fara@ubuntu:~$ sudo docker exec -it single-node-wazuh.manager-1 bash
wazuh.manager-1 bash
bash-5.2# grep -A 10 "vulnerability-detection" /var/ossec/etc/ossec.conf
<vulnerability-detection>
  <enabled>yes</enabled>
  <index-status>yes</index-status>
  <feed-update-interval>60m</feed-update-interval>
</vulnerability-detection>

<indexer>
  <enabled>yes</enabled>
  <hosts>
    <host>https://wazuh.indexer:9200</host>
  </hosts>
  <ssl>
    <certificate_authorities>
      <ca>/etc/ssl/root-ca.pem</ca>
    </certificate_authorities>
  </ssl>
</indexer>
bash-5.2#
```

The module successfully initialized and began downloading its CVE feed database, confirmed through consistent growth of the feed storage directory over time (1.4GB → 5GB+ during the assessment window), indicating an actively progressing, legitimate download of a large vulnerability dataset covering multiple operating systems and package ecosystems.

Vulnerability Findings — Triage Table

CVE	Severity	Affected Host	CVSS	Recommended Remediation
CVE-2024-0727	High	agent-ubuntu:server	7.5	Update the cryptography package to the latest patched version .
CVE-2023-41105	High	agent-ubuntu:server	7.1	Upgrade Python packages to the latest security-patched version provided by Ubuntu.
CVE-2024-28948	Medium	agent-ubuntu:server	4	Update snapd to the latest available version.
CVE-2024-43810	Medium	agent-ubuntu:server	5.3	Upgrade the Twisted package to the latest fixed release.
CVE-2016-2781	Low	agent-ubuntu:server	2.1	Update the coreutils package to the latest supported version.

Part 3 — Custom SOC Dashboard

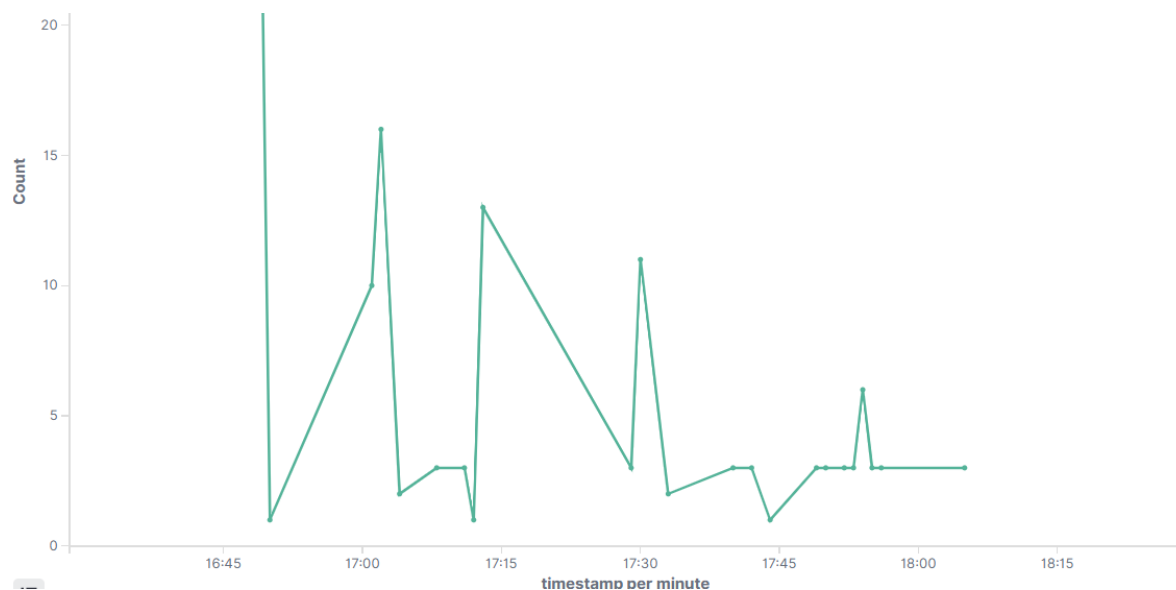
1. Top 10 Alert Types

A data table visualization built against the `wazuh-alerts-*` index, using a Terms aggregation on `rule.description` to display the 10 most frequent alert types across all agents. Results included routine activity

rule.description: Descending	Count
Host-based anomaly detection event (rootcheck).	34
PAM: Login session closed.	20
PAM: Login session opened.	20
Successful sudo to ROOT executed.	15
Dpkg (Debian Package) half configured.	9
New dpkg (Debian Package) installed.	6
Wazuh agent started.	5
Auditd: Device enables promiscuous mode.	3
Log file rotated.	3
Wazuh agent stopped.	3
10	10

2. Alerts Over Time

A line chart using a Date Histogram aggregation on the `timestamp` field, visualizing alert volume trends across the monitoring period.



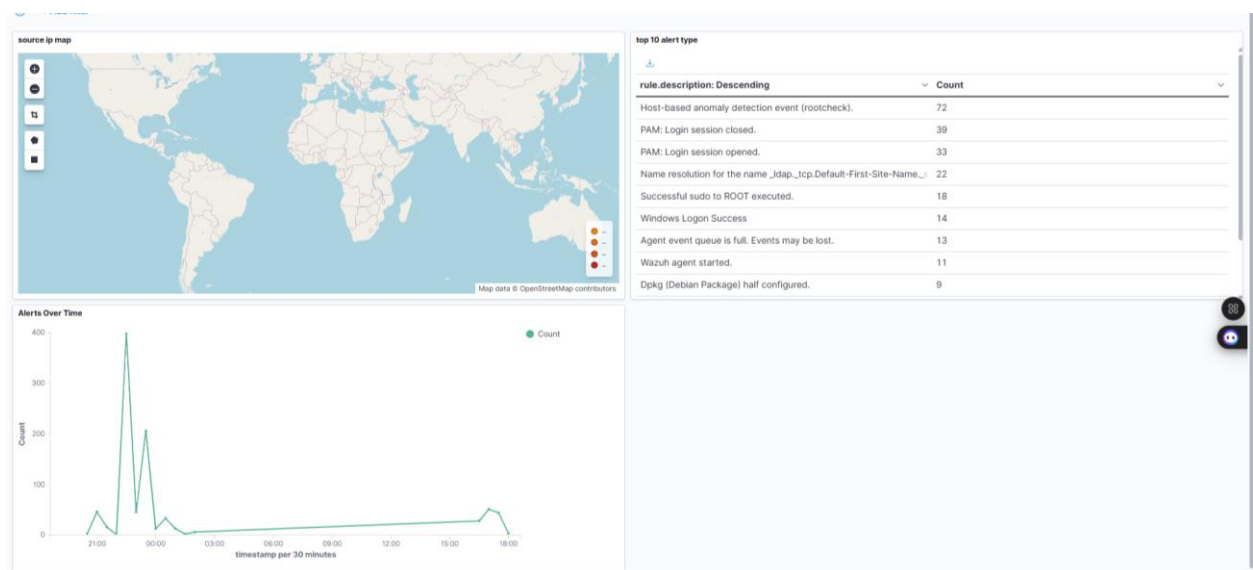
3. Source IP World Map

A coordinate map visualization configured with Geohash aggregation to plot the geographic origin of alert source IPs. The map rendered correctly but displayed no data points, as all lab traffic originates from private/internal IP addresses (192.168.56.x range), which have no real-world geographic mapping in GeoIP databases. This is expected and correct behavior for an isolated lab environment; in a production deployment monitoring internet-facing systems, this visualization would display genuine attacker geolocation data.

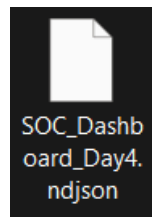


Dashboard Assembly and Export

All three visualizations were combined into a single dashboard named **"SOC Dashboard - Day 4"** and exported via Stack Management → Saved Objects as an NDJSON file — OpenSearch Dashboards' standard JSON-based export format for saved objects, suitable for import into other Wazuh deployments or delivery to a client as a SOC dashboard artefact.



Deliverable file: SOC_Dashboard_Day4.ndjson



Conclusion

Day 4 successfully demonstrated Wazuh's File Integrity Monitoring capability through a controlled test on `/etc/passwd`, confirming real-time detection of unauthorized file changes. The Vulnerability Detector was verified as correctly enabled and actively processing its CVE feed database, with findings to be finalized upon completion of the initial download. A custom three-panel SOC dashboard was built and exported as a portable NDJSON artefact, demonstrating the ability to translate raw security data into client-ready visual reporting — directly reflecting real MSSP deliverable standards.